

Acute Dyspnea & Hypoxemia — Bedside Reference

Differential by system, then one number — the A-a gradient — splits hypoxemia in half. Adapted from TeachIM.org (Zhang, Dec 2022), CC BY-NC 4.0.

Dyspnea differential by system

System	Causes
Resp — airway	Anaphylaxis/angioedema, mucus plug, stenosis, bronchospasm (COPD/asthma)
Resp — alveoli	Edema, PNA, DAH, atelectasis, pleural process
Resp — vessel	PE
Cardiac	MI, arrhythmia, valvular, pericardial/tamponade
MSK	Diaphragm dysfunction, abdominal pressure
3 A's	Anemia, anxiety, acidosis

Resp + cardiac are most common. Toolbox: **CXR, blood gas, CBC, ECG** + POCUS.

A-a gradient

$PAO_2 = FiO_2 \times (Patm - PH_2O) - PaCO_2 / 0.8 \approx 150 - 1.25 \times PaCO_2$ (room air)

$A-a = PAO_2 - PaO_2$; expected = **age/4 + 4** (64 yo → ~20)

Worked: PaO_2 50, $PaCO_2$ 32 → PAO_2 110; A-a = 60 (elevated vs expected 20).

5 causes of hypoxemia

Cause	A-a	Note
Low FiO_2	Normal	Altitude; not acute
Hypoventilation	Normal	$\uparrow PaCO_2$; opioids/CNS/NM weakness
V/Q mismatch	$\uparrow$	Most common; PE, obstruction
Shunt	$\uparrow$	No correction w/ 100% O_2
$\downarrow$ Diffusion	$\uparrow$	Chronic (ILD, emphysema)

Algorithm: normal A-a → hypoventilation (check $PaCO_2$); elevated A-a → V/Q vs shunt (100% O_2 challenge).

ABG vs VBG

Get an **ABG when you need the PaO_2** : confirm/grade hypoxemia or compute the A-a gradient. Else VBG suffices.

$ABG\ pH \approx VBG\ pH + 0.04$; $ABG\ PaCO_2 \approx VBG\ PvCO_2 - 4$.

Hypoxia that isn't hypoxemia (normal PaO_2)

- Anemic; stagnant (low flow / cardiac)
- Histotoxic — cyanide (sodium nitroprusside)
- Methemoglobinemia — nitrates, local anesthetics, sulfonamides
- SpO_2 falsely reassuring in CO poisoning / MetHb → confirm with co-oximetry

POCUS / pearls

- BLUE-protocol lung US: B-lines (edema), consolidation (PNA), absent sliding (PTX), effusion — high sensitivity
- Dyspnea ≠ hypoxemia; normal PaO_2 75–100
- Shunt = perfusion without ventilation; fails the 100% O_2 challenge
- Tamponade clues: hypotension, $\uparrow$ JVP, electrical alternans, enlarged cardiac silhouette